AnalogForge Hand Work Guide

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Purpose

This guide turns the Analog Forge kit into a start-to-finish hand workflow for exploring a novel, speculative, or theoretical science question. It is written for work that begins before the math is settled.

The rule is simple:

```
make the state visible
-> place C
-> test the boundary
-> split proof from obligation
-> receipt the result
-> continue or start a new page
```

The kit is not decoration. It is the physical state machine that mirrors the digital ForgeFactory workbench.

Minimum Workstation

Use these minimum pieces:

Piece | Role

Grey page | unresolved starting substrate
Red, green, blue marks | active triadic state read
White mark/page | proof or visible continuation
Black mark/page | obligation or unresolved remainder
Clear sleeve | reversible test layer
C token | current local center
Strings | dependencies, braids, edges, boundary crossings
Notecards | decision probes and possible routes
Cards | ordering, witness, permutation, exception
Dice | bounded uncertainty only
Receipt sheet | closure and recovery record

Start-To-Finish Run

1. Name the question.

Write one sentence on a grey page:

```
What is the phenomenon, claim, object, or system being explored?
```

Do not start with a theory. Start with the observable thing.

2. Place C.

Put a token, sticker, circle, or card at the current center. C is the local thing being read now. Every other object is interpreted relative to this center.

3. Add the first three-color gradient.

Every live page needs at least three colors. Use them as:

```
active state / carrier state / boundary state
```

or:

```
left / center / right
```

or:

```
known / current / unknown
```

4. Enumerate what is physically visible.

Make a short list:

```
objects
forces
constraints
cycles
failures
symmetries
weird exceptions
measurements
unknowns
```

Each item gets a token, sticker, card, or notecard.

5. Ask the base decision.

For each item, ask:

```
Does this continue on the current page?
or
Does this require a new page?
```

If it continues, keep it near C.
If it opens a new state, start a new grey page and string it back to the source.

6. Split white and black.

Mark every result:

```
white = proof-bearing continuation
black = obligation, unresolved remainder, shadow branch
```

Never delete black material. Preserve it as future work.

7. Bind only after testing.

A page can bind into a notebook or binder only when its relation is clear:

```
same color
conjugate color
gradient-linked
carrier-linked
boundary-authorized
explicitly receipted
```

If the relation cannot be stated, leave the page loose.

8. Write the receipt.

Record:

```
question
C position
colors
tools used
boundary touched
decision made
proof continuation
black obligation
next page or binder
```

The receipt is what makes the hand run recoverable.

Boundary Collision Protocol

Use this whenever two states collide, contradict, overlap, fail, or force a new interpretation.

1. Stop movement.

Do not merge the pages immediately.

2. Mark the collision in neon.

Neon means active boundary. Put it directly on the collision point or on a clear overlay above the collision.

3. Place a boundary card.

Write:

```
What touched?
What changed?
What cannot pass?
What must be conserved?
What new page may be born?
```

4. Split the collision into four reads.

Read | Physical mark

incoming state | colored token/string entering boundary
boundary surface | neon line or clear overlay mark
reflected state | token/string returning to source side
transmitted state | token/string crossing to new page

5. Test conservation.

Ask:

```
Did any color disappear?
Did any obligation disappear?
Did any string lose its source or target?
Did any token move without a receipt?
```

If yes, create a black obligation receipt.

6. Decide collision result.

Choose one:

```
pass through
reflect
split
bind
branch to new page
stay unresolved
```

7. Receipt the collision.

Every collision receipt needs:

```
source page
target page
C before
C after
boundary mark
incoming colors
outgoing colors
white continuation
black obligation
```

```
next action
```

Exploring A Novel Or Theoretical Science

Use this when the subject is not yet stable enough for normal equations.

1. Build the observation field.

Put every observed feature on separate notecards. Do not rank them yet.

2. Pick a first C.

Choose the feature that everything else seems to orbit, constrain, disturb, or explain.

3. Make three piles.

```
known
unknown
boundary-active
```

Mark boundary-active with neon.

4. Build the first relation graph.

Use strings:

```
straight string = direct dependency
looped string = feedback
knotted string = unresolved dependency
braided strings = interacting routes
black string = obligation relation
white string = proof continuation
```

5. Run the eight-neighbor check.

Around C, reserve eight positions:

```
N, NE, E, SE, S, SW, W, NW
```

Place the strongest adjacent features in those slots. Empty slots become black obligations or future measurement targets.

6. Create candidate laws.

Each candidate law gets a notecard:

```
When X crosses boundary Y, state Z changes by rule R.
```

Do not call it true yet.

7. Stress the candidate law.

Use:

```
card spread = ordering/permutation stress
dice = bounded uncertainty stress
clear overlay = reversible alternate route
black page = failure case
white page = passed case
```

8. Convert failures into obligations.

Every failed case must say what would make it resolvable:

```
new measurement
new material
new model
new boundary condition
new color relation
```

```
new page
```

9. Produce a research map.

At the end of the run, the binder should contain:

```
observation pages
C history
boundary collisions
candidate laws
proof continuations
obligations
receipts
replacement/material notes
next experiment list
```

What To Do At Boundary Collisions

If the state breaks:

```
mark neon
freeze the current C
copy the pre-collision page
move the copy into clear overlay
split white/black
receipt both sides
```

If two explanations compete:

```
put each explanation on its own grey page
connect both to the same C
use card order to test sequence
use black cards for contradictions
use white cards for shared continuations
```

If an object belongs to two states:

```
use a clear sleeve
place one color on the base page
place the second color on the overlay
bind only after a receipt explains the relation
```

If the question becomes too large:

```
create a new binder
copy the source receipt
start a new C
leave a string back to the parent binder
```

Completion Rule

A hand run is complete only when it has:

```
one named C history
at least one receipt
white continuation or black obligation
known next action
archive location
```

If it has no receipt, it is still scratch work.